Intelligent Sensors
H. Yamasaki (Editor)

What are the Intelligent Sensors

Hiro YAMASAKI

Yokogawa Research Institute Corporation, Musashino, Tokyo, Japan

(Professor Emeritus of The University of Tokyo)

1. INTRODUCTION

Sensors incorporated with dedicated signal processing functions are called intelligent sensors or smart sensors. Progress in the development of the intelligent sensors is a typical example of sensing technology innovation. The roles of the dedicated signal processing functions are to enhance design flexibility of sensing devices and realize new sensing functions. Additional roles are to reduce loads on central processing units and signal transmission lines by distributing information processing in the sensing systems [1–5].

Rapid progress in measurement and control technology is widely recognized. Sensors and machine intelligence enhance and improve the functions of automatic control systems. Development of new sensor devices and related information processing supports this progress.

Application fields of measurement and control are being rapidly expanded. Typical examples of newly developed fields are robotics, environmental measurement and biomedical areas. The expanded application of intelligent robots requires advanced sensors as a replacement of the sensory systems of human operators. Biomedical sensing and diagnostic systems are also promising areas in which technological innovation has been triggered by modern measurement and automatic control techniques.

Environmental instrumentation using remote sensing systems, on the other hand, informs us of an environmental crisis on the earth.

I believe that in these new fields attractive functions and advanced features are mostly realized by intelligent sensors. Another powerful approach to advanced performance is the systematization of sensors for the enhancement of sensing functions.

In this chapter the recent development of intelligent sensors and their background is described.

2. WHY INTELLIGENT SENSORS?

2.1 General information flow through three different worlds

Measurements, control and man–machine communications

Let us consider the technical background of the intelligent sensors. First we will analyze the general information flow relating to measurement and control systems. This information flow includes not only communications between objects and measurement systems, but also man–machine communications. It can be depicted as communication between three different worlds, (Fig.1). It is easily understood that a smooth and efficient information flow is essential to make the system useful and friendly to man [6].

The three different worlds shown in Figure 1 are as follows:

(1) Physical world. It represents the measured and controlled object, sensors and actuators. Natural laws rule this world in which the causality is strictly established. Information is transmitted as a physical signal.

(2) Logical world. It represents an information processing system for measurement and control. Logic rules this world in which information is described by logical codes.

(3) Human intellectual world. It is the internal mental world of the human brain. Information is translated to knowledge and concepts in this world. What kinds of laws rule this world? The author cannot answer this question at the moment. This is the one of the old problems left unsolved.

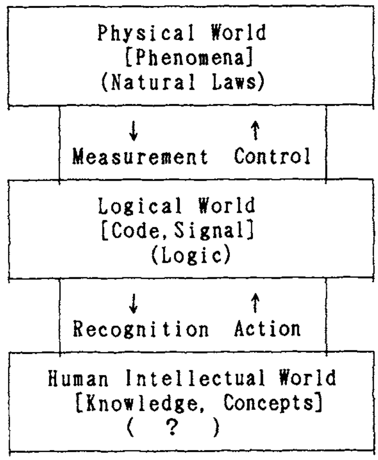

Figure 1 Information flow between three different worlds

Information in the physical world is extracted by sensing or measurement techniques and is transferred into logical world. The information is processed according to the objectives of the measurement and control systems in the logical world, and it is recognized by man through man–machine interfaces that display the measured and processed results. Thus, the information is transferred into the human intellectual world and then constructed as a part of the knowledge and concepts in the human world. Masses of obtained information segments are first structured and then become a part of knowledge. Assembled knowledge constitutes concepts. Groups of concepts constitute a field of science and technology.

Human behavior is generally based on obtained information and knowledge. Intentions are expressed as commands and transmitted to the system by actions toward man–machine interfaces. The logical control system regulates its objects through actuators in the physical world based on the command. We can paraphrase "measurement and control" as the information flow process: measurement, recognition, action and control in three different worlds in which human desire is realized in the physical world.

Between the physical world and the logical one, information is communicated through measurement systems and control systems. There, sensors and actuators act as the interfaces. Sensors are the front-end elements of the sensing or measuring systems, and they extract information from the objects in the physical world and transmit the physical signal.

Between the logical world and the human intellectual world, the communications are carried out through a man-machine interface.

An intelligent sensor is one which shifts the border of the logical world to to the physical world. Part of the information processing of the logical world is replaced by information processing within the sensors.

An intelligent man-machine interface comprises a system by which the border between the human world and the logical world is shifted to the human side. Part of human intelligence for recognition is replaced by machine intelligence of the man-machine interfaces.

2.2 Needs for intelligent sensors

Neede for present sensor technology

Present sensor technology has been developed to meet the various past needs of sensors. Sensor technology is essentially needs-oriented. Therefore, we can forecast the future of sensing technology by discussing the present needs of sensors.

The present needs for the development of intelligent sensors and sensing systems can be described as follows: [6,7]

(1) We can sense the physical parameters of objects at normal states with high accuracy and sensitivity. However, detection of abnormalities and malfunctions are poorly developed. Needs for fault detection and prediction are widely recognized.

(2) Present sensing technologies can accurately sense physical or chemical quantities at single point. However, sensing of multidimensional states is difficult. We can imagine the environment as an example of a multidimensional state. It is a widely spread state of characteristic parameters are spatially dependent and time-dependent.

(3) Well-defined physical quantities can be sensed with accuracy and sensitivity. However, quantities that are recognized only by human sensory systems and are not clearly defined cannot be sensed by simple present sensors. Typical examples of such quantities are olfactory odors and tastes.

Difficulties in the above-described items share a common feature: the ambiguity in definition of the quantity or difficulty in simple model building of the objects. If we can define the object clearly and build its precise model, we can learn the characteristic parameter that describes the object's state definitely. Then, we can select the most suitable sensor for the characteristic parameter. In the above three cases, this approach is not possible.

Human expertise can identify the abnormal state of an object. Sometimes man can forecast the trouble utilizing his knowledge of the object. Man and animals know whether their environment is safe and comfortable because of their multimodal sensory information. We can detect or identify the difference in complicated odors or tastes in foods. Man senses these items without the precise model of the object by combining multimodal sensory information and knowledge. His intelligence combines this multiple sensory information with his knowledge.

Accordingly in the world of physical sensors, we can solve these problems by signal processing and knowledge processing. Signal processing combines or integrates the outputs of multiple sensors or different kinds of sensors utilizing knowledge of the object. This processing is called sensor signal fusion or integration and is a powerful approach in the intelligent sensing system. An advanced information processing for sensor signal fusion or integration should be developed and is discussed later in another chapter.

3. ARCHITECTURE OF INTELLIGENT SENSING SYSTEMS

3.1 Hierarchy of machine intelligence [8]

Before we describe intelligent sensors, we will discuss the architecture of intelligent sensing systems in which intelligent sensors are used as the basic components. Flexibility and adaptability are essential features in an intelligent sensing system. We can consider the human sensory system as a highly advanced example of an intelligent sensing system. Its sensitivity and selectivity are adjustable corresponding to the objects and environment.

The human sensory system has advanced sensing functions and a hierarchical structure. This hierarchical structure is suitable architecture for a complex system that executes advanced functions.

Upper layer [KNOWLEDGE PROCESSING] TOTAL CONTROL Concentrated central processing (Digital serial processing)
Middle layer [INFORMATION PROCESSING] INTERMEDIATE CONTROL, TUNING & OPTIMIZATION OF LOWER LEVEL SENSOR SIGNAL INTEGRATION & FUSION
Lower layer [SIGNAL PROCESSING] SENSING & SIGNAL CONDITIONING [INTELLIGENT SENSORS] Distributed parallel processing (Analog)

Figure 2 Hierarchical structure of intelligent sensing system

One example of the hierarchical architecture of intelligent sensing systems has a multilayer as shown in Figure 2.

The most highly intelligent information processing occurs in the top layer. Functions of the processing are centralized, like in the human brain. Processed information is abstract and independent of the operating principle and physical structure of sensors.

On the other hand, various groups of sensors in the bottom layer collect information from external objects, like our distributed sensory organs. Signal processing of these sensors is conducted in a distributed and parallel manner. Processed information is strongly dependent on the sensor's principles and structures

Sensors incorporated with dedicated signal processing functions are called intelligent sensors or smart sensors. The main roles of dedicated signal processing are to enhance design flexibility and realize new sensing functions. Additional roles are to reduce loads on central processing units and signal transmission lines by distributing information processing in the lower layer of the system.

There are intermediate signal processing functions in the middle layer. One of the intermediate signal processing functions is the integration of signals from multiple sensors in the lower layer. When the signals come from different types of sensors, the function is referred to as sensor signal fusion. Tuning of sensor parameters to optimize the total system performance is another function.

In general, the property of information processing done in each layer is more directly oriented to hardware structure in the lower layer and less hardware-oriented in the upper layer. For the same reason, algorithms of information processing are more flexible and need more knowledge in the upper layer and are less flexible and need less knowledge in the lower layer.

We can characterize processing in each layer as follows: signal processing in the lower layer, information processing in the middle layer, and knowledge processing in the upper layer.

3.2 Machine intelligence in man-machine interfaces

Intelligent sensing systems have two different types of interfaces. The first is an object interface including sensors and actuators. The second is a man-machine interface. The man-machine interface has a hierarchical structure like the one as shown in Figure 2. Intelligent transducers are in the lower layer, while an advanced function for integration and fusion is in the middle layer. Machine intelligence in the middle layer harmonizes the differences in information structure between man and machine to reduce human mental load. It modifies the display format and describes an abnormal state by voice alarm. It can also check error in human operation logically, which makes the system more user friendly.

3.3 Role of intelligence in sensors

Sensor intelligence performs a distributed signal processing in the lower layer of the sensing system hierarchy. The role of the signal processing function in intelligent sensors can be summarized as follows:

1) reinforcement of inherent characteristics of the sensor device and
2) signal enhancement for the extraction of useful features of the objects.

1) Reinforcement of inherent characteristics of sensor devices

The most popular operation of reinforcement is compensation of characteristics, the suppression of the influence of undesired variables on the measurand. This operation of compensation is described below. We can depict the output signal of a sensor device by the eqation

$$F(x_1,\ x_2 \ldots x_n),$$

where x_1 is the quantity to be sensed or measured and $x_2 \ldots x_n$ are the quantities which affect the sensor output or sensor performances. The desirable sensor output is dependent only on x_1 and t and independent of any change in $x_2 \ldots x_n$. Compensation is done by other sensors that are sensitive to change in $x_2 \ldots x_n$. A typical compensation operation can be described mathematically for the case of a single influential variable for simplicity. The increment of x_1 and x_2 are represented by Δx_1 and Δx_2, espectively. The resultant output y is as follows:

$$y = F(x_1+\Delta x_1,\ x_2+\Delta x_2) - F^*(x_2+\Delta x_2) \tag{1}$$

$$\begin{aligned} y \approx\ & F(x_1,\ x_2) + (\partial F/\partial x_1)\Delta x_1 + (\partial F/\partial x_2)\Delta x_2 \\ & + \tfrac{1}{2}\{(\partial^2 F/\partial x_1^2)(\Delta x_1)^2 + 2(\partial^2 F/\partial x_1 \partial x_2)\Delta x_1 \Delta x_2 + (\partial^2 F/\partial x_2^2)(\Delta x_2)^2\} \\ & - F^*(x_1,\ x_2) - (\partial F^*/\partial x_2)\Delta x_2 - \tfrac{1}{2}(\partial^2 F^*/\partial x_2^2)(\Delta x_2)^2 \end{aligned} \tag{1'}$$

If we can select function F^* to satisfy the conditions in eq. (3) within variable range of x_2, we obtain eq. (2), and can thus reduce the influence of change in x_2, as shown in Figure 3.

$$y = (\partial F/\partial x_1)\Delta x_1 + \tfrac{1}{2}(\partial^2 F/\partial x_1^2)(\Delta x_1)^2 + (\partial^2 F/\partial x_1 \partial x_2)\Delta x_1 \Delta x_2 \tag{2}$$

$$\begin{aligned} F(x_1, x_2) &= F^*(x_1, x_2) \\ (\partial F/\partial x_2) &= (\partial F^*/\partial x_2) \\ (\partial^2 F/\partial x_2^2) &= (\partial^2 F^*/\partial x_2^2) \end{aligned} \tag{3}$$

However, since eq. (2) contains a second-order cross-term including x_2, the compensation is not perfect. In this case, the output at the zero point cannot be influenced by x_2, but the output at the full scale point of x_1 is influenced by x_2. When $F(x_1, x_2)$ is expressed as a linear combination of single variable functions, as shown in eq. (4), then the cross-term is zero and the compensation is perfect.

$$F(x_1, x_2) = \alpha F_1(x_1) + \beta F_2(x_2) \tag{4}$$

When $F(x_1, x_2)$ is expressed as a product of single variable functions, as shown in eq. (5), we can realize the perfect compensation by dividing $F(x_1, x_2)$ by the suitable function $F_2(x_2)$.

$$F(x_1, x_2) = \gamma \cdot F_1(x_1) \cdot F_2(x_2) \tag{5}$$

Conventional analog compensation is usually represented by eq. (2); thus perfect compensation is not realized easily. If we can use the computer for the operation of compensation, we can obtain much more freedom to realize perfect compensation. We can suppress the influence of undesired variables using table-lookup method as long as the undesired variables can be measured, even if the relationship between input and output is complicated. Therefore, the new sensor device can be used. But if the sensitivity for the undesired variables cannot be measured, the device cannot be used.

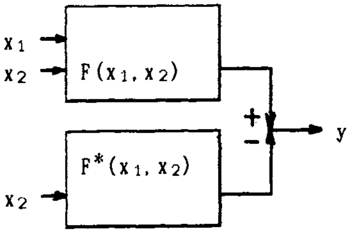

Figure 3 Fundamental structure of compensation for undesired variables

The compensation technique described above is considered to be a static approach to the selectivity improvement of the signal. This approach takes advantage of the difference in the static performance of the sensor device for the signal and the undesired variables or noise. Compensation is the most fundamental role for machine intelligence in relation to intelligent sensors.

2) Signal enhancement for useful feature extraction

Various signal processing is done for useful feature extraction. The goals of signal processing are to eliminate noise and to make the feature clear.

Signal-processing techniques mostly utilize the differences in dynamic responses to signal and noise, and are divided into two different types, frequency-domain processing and time-domain processing. We consider them a dynamic approach for selectivity improvement of the signal (Fig. 4).

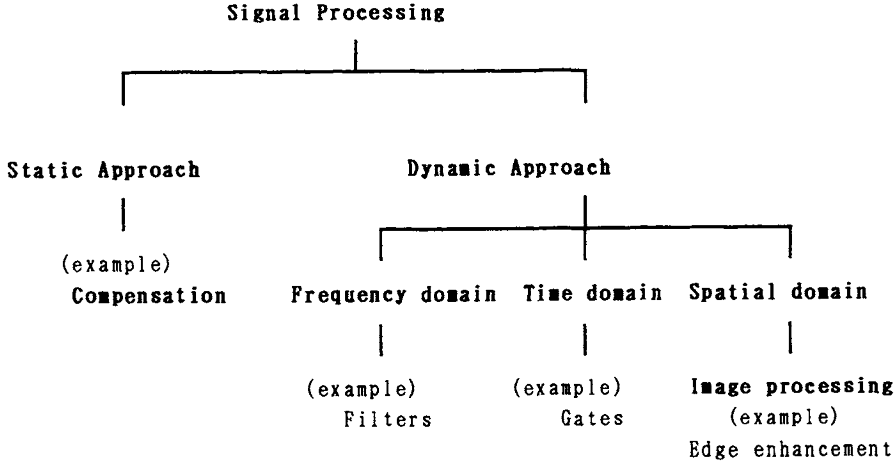

Figure 4 A classification of signal processing relating to intelligent sensors

The typical signal processing used in intelligent sensors is real–time filtering of noise. They are low–pass, high–pass and band–pass filtering in frequency–domain processing. The filtering is done mostly in analog circuits.

Another important area of signal processing in intelligent sensors is the spatial domain. The most popular spatial domain processing is image processing for visual signal. Noise suppression, averaging, and edge enhancement are functions of processing, and primary features of the target image are extracted.

More advanced image processing, for example for pattern recognition, is usually done in a higher hierarchical layer. However, information quantity of the visual signal from multipixel image sensors is clearly enormous, and primary image processing is very useful for the reduction of load for signal transmission line and processors in higher layers.

We can summarize the roles of intelligence in sensors as follows:

The most important role of sensor intelligence is to improve signal selectivity of individual sensor devices in the physical world. This includes simple operations of output from multiple sensor devices for primary feature extraction, such as primary image processing. However, this does not include optimization of device parameters or signal integration from multiple sensor devices, because this requires knowledge of the sensor devices and their object.

3.4 Roles of intelligence in middle layer

The role of intelligence in the middle layer is to organize multiple output from the lower layer and to generate intermediate output. The most important role is to extract the essential feature of the object. In the processing system in the middle layer, the output signals from multiple sensors are combined or integrated. The extracted features are then utilized by upper layer intelligence to recognize the situation. This processing is conducted in the logical world.

<u>Sensor signal fusion and integration</u> [8]

We can use sensor signal integration and sensor signal fusion as the basic architecture to design an adaptive intelligent sensing system. Signals from the sensor for different measurands are combined in the middle layer and the results give us new useful information. Ambiguity or imperfection in the signal of a measurand can be compensated for by another measurand. This processing creates a new phase of information.

Optimization

Another important function of the middle layer is parameter tuning of the sensors to optimize the total system performance. Optimization is done based on the extracted feature and knowledge of target signal. The knowledge comes from the higher layer as a form of optimization algorithm.

Example of architecture of intelligent sensing system:"Intelligent microphone"

Takahashi and Yamasaki [9] have developed an intelligent adaptive sound-sensing system (Fig. 5), called the "Intelligent Microphone", this is an example of sensor fusion of auditory and visual signals. The system receives the necessary sound from a signal source in various noisy environments with improved S/N ratio. The location of the target sound source is not given, but some feature of the target is a cue for discriminating the signal from the noise. The cue signal is based on the visual signal relating to the movement of the sound-generating target. The system consists of an audio subsystem and a visual subsystem. The audio subsystem consists of one set of multiple microphones, a multiple input linear filter and a self-learning system for adaptive filtering. The adaptive sensing system has a three-layered structure, as shown in Figure 6. The lower layer includes multiple microphones and A/D converters. The middle layer has a multiple input linear filter, which combines multiple sensor signals and generates an integrated output. A computer in the upper layer tunes and controls the performance of the filter. The middle layer filter functions as an advanced signal processor for the middle layer, as described previously. The computer in the upper layer has a self-learning function and optimizes filter performance on the basis of knowledge. The knowledge is given by an external source (in this case, the source is man).

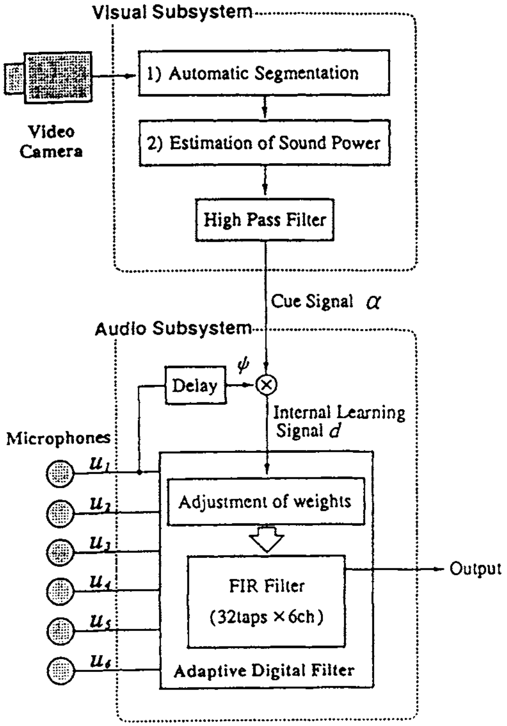

Figure 5 Block diagram of the intelligent sound-sensing system consisting of an audio subsystem and a visual subsystem

The visual subsystem consists of a visual sensor and image-sensing system. It also has three layers in its hierarchical structure. It extracts the object's movement and generates the cue signal. The output from the optimized filter is the final output of the system. Max S/N improvement of 18 dB has been obtained experimentally.

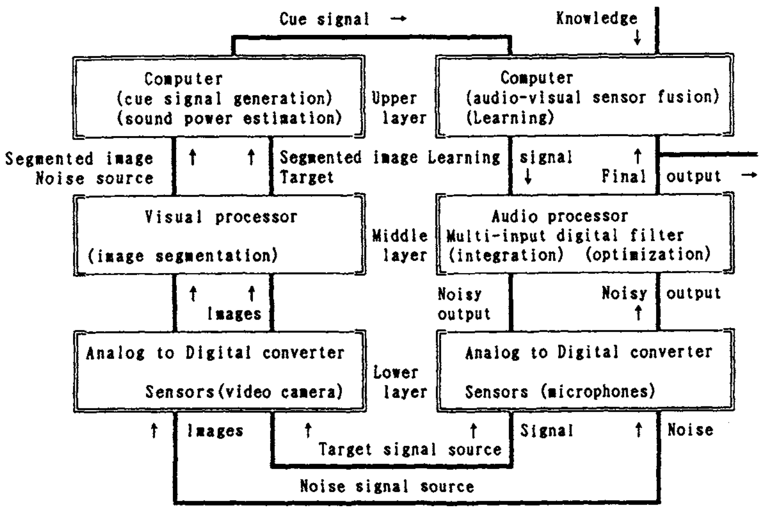

Figure 6 Intelligent adaptive sound sensing system.
Both subsystems have a 3-layer hierarchical structure

Possible applications of the system are the detection of human voice or abnormal sounds from failed machines in a noisy environment.

4. APPROACHES TO SENSOR INTELLIGENCE REALIZATION

In the author's opinion, there are three different approaches to realize sensor intelligence ([10] and Fig. 7).

1) Integration with computers
   (Intelligent integration)
2) Use of specific functional materials
   (Intelligent materials)
3) Use of functional geometrical structure
   (Intelligent structure)

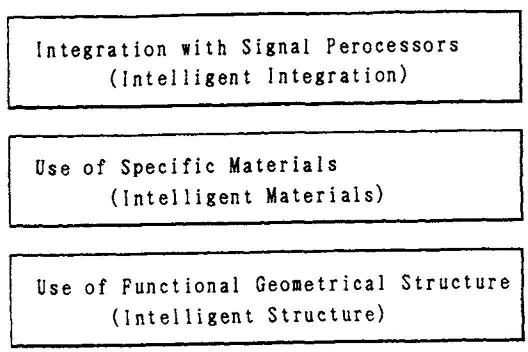

Figure 7 Three different approaches to realize sensor intelligence

The first approach is the most popular. A typical example of intelligent integration is integration of the sensing function with the information processing function. It is usually imaged as the integration of sensor devices and microprocessors. The algorithm is programmable and can be changed even after the design has been made.

Information processing is optimization, simple feature extraction, etc. Some operation is done in real time and in parallel with analog circuitry or network. We call such integration a computational sensor [11].

In the second and third approaches, signal processing is analog signal discrimination. Only the useful signal is selected and noise or undesirable effects are suppressed by the specific materials or specific mechanical structure. Thus, signal selectivity is enhanced.

In the second approach, the unique combination of object material and sensor material contributes to realization of an almost ideal signal selectivity. Typical examples of sensor materials are enzymes fixed on the tip of biosensors.

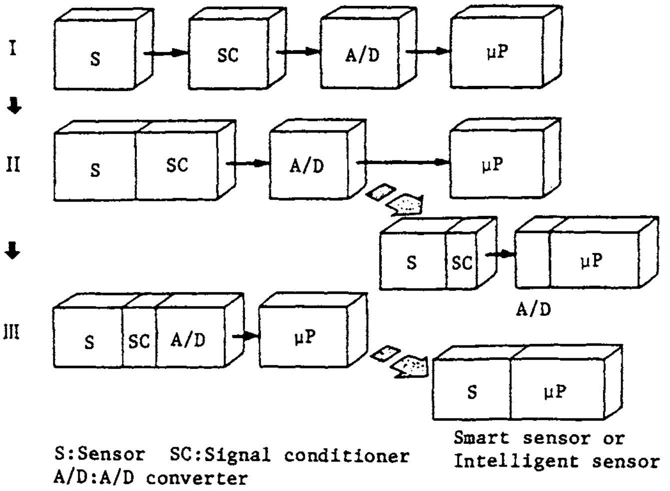

Figure 8 Trends in integration with microprocessors and sensors

In the third approach, the signal processing function is realized through the geometrical or mechanical structure of the sensor devices. Propagation of optical and acoustic waves can be controlled by the specific shape of the boundary between the different media. Diffraction and reflection of the waves are controlled by the surface shape of the reflector. A lens or a concave mirror is a simple example. Only light emitted from a certain point in the object space can be concentrated at a certain point in the image space. As a result the effect of stray light can be rejected on the image plane.

The hardware for these analog processes is relatively simple and reliable and processing time is very short due to the intrinsically complete parallel processing. However, the algorithm of analog processing is usually not programmable and is difficult to modify once it is fabricated.

Typical examples of these three technical approaches are described in the following sections. They range from single-chip sensing devices integrated with microprocessors to big sensor arrays utilizing synthetic aperture techniques, and from two-dimensional functional materials to a complex sensor network system.

4.1 Approach using integration with computer

The most popular image of an intelligent sensor is an integrated monolithic device combining sensor with microcomputer within one chip. The development process toward such intelligent sensors is illustrated in Figure 8 [12]. Four separate functional blocks (sensor, signal conditioner, A/D converter and microprocessor) are gradually coupled on a single chip, then turned into a direct coupling of sensor and microprocessor. However, such a final device is not yet in practical use.

We are in the second or third stage of Figure 8. Many different examples are introduced by the other authors in this book. Below some of the examples in the stages will be discussed; for example, two samples of pressure sensors that are in practical use for process instrumentation. Figure 9 shows a single–chip device including a silicon diaphragm differential pressure sensor, an absolute pressure sensor, and a temperature sensor. The chip is fabricated according to a micromachining technique. The output signals from these three sensors are applied to a microprocessor via an A/D converter on a separate chip. The processor calculates the output and at the same time compensates for the effects of absolute pressure and temperature numerically. The compensation data of each sensor chip is measured in the manufacturing process and loaded in the ROM of the processor, respectively. Thus, the compen–sation is precise and an accuracy of 0.1% is obtained as a differential pressure transmitter [13].

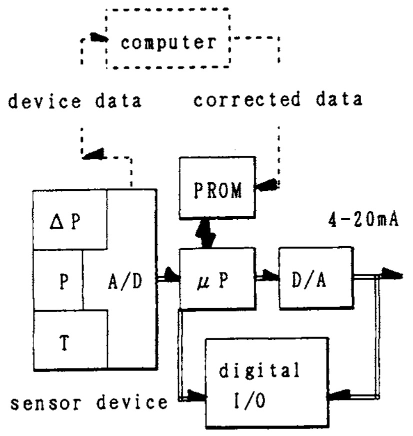

Figure 9 A smart differential pressure transmitter with digital signal compensation

Figure 10 shows another pressure sensor, which has a silicon diaphragm and a strain–sensitive resonant structure also made by micromachining technology. Two "H" shape vibrators are fabricated on a silicon pressure–sensing diaphragm as shown in the figure, and they vibrate at their natural frequencies, incorporating oscillating circuits. When differential pressure is applied across the diaphragm, the oscillation frequency of one unit is increased and the other one is decreased due to mechanical deformation of the diaphragm. The difference frequency between two oscillators is proportional to the differential pressure. The mechanical vibrators are sealed inside a local vacuum shell to realize a high resonant Q factor. The effects of change in temperature and static pressure are automatically canceled by the differential construction. Sensors with a frequency output are advantageous in interfacing with microprocessors.

These differential pressure transmitters have a pulse communication ability that is superposed on the analog signal line by a digital communication interface. Remote adjustment of span and zero, remote diagnosis and other maintenance functions can be performed by digital communication means.

The range of analog output signal is the IEC standard of 4–20 mA DC. Therefore, total circuits, including the microprocessor, should work within 4 mA. The problem can be overcome by a CMOS circuit approach [14].

4.2 Approach using specific functional materials

Enzymes and microbes have a high selectivity for a specified substance. They can even recognize a specific molecule. Therefore, we can minimize the time for signal processing by rejecting the effects of coexisting chemical components.

One example of an enzyme sensor is the glucose sensor. Glucose–oxidase oxidizes glucose exclusively and produces gluconic acid and hydrogen peroxide (H_2O_2). An electrode detecting H_2O_2 generates an electric current that is proportional to the glucose concentration. The enzyme is immobilized on the sensor tip. In this way, we can develop a variety of biosensors

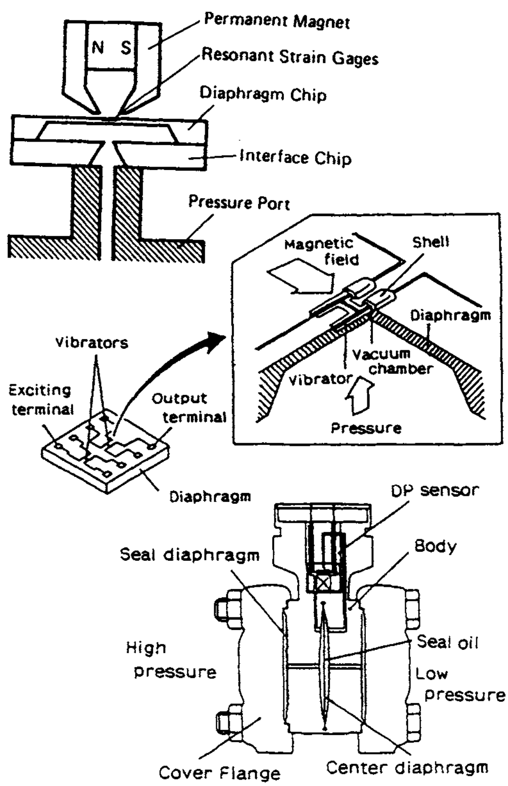

Figure 10 A differential pressure sensor having a strain–sensitive resonant structure fabricated by micromachining

utilizing various enzymes and microbes. If we immobilize the antigen or antibody on the sensor tip, we can achieve a high sensitivity and selectivity of the immuno assay. This type of sensor is called a biosensor because it uses the biological function of enzymes. Since biosensors have almost perfect selectivity, to detect many kinds of species the same amounts of different biosensors are necessary.

Another approach to the chemical intelligent sensor for various species has been proposed. It uses multiple sensors with different characteristics and imperfect selectivity. Kaneyasu describes an example of the approach in the chapter on the olfactory system. Six thick film gas sensors are made of different sensing material, which have different sensitivities for the various object gases. They are mounted on a common substrate and the sensitivity patterns of the six sensors for the various gases are recognized by microcomputer. Several examples of sensitivity (conductivity) patterns for organic and inorganic gases have been found. Typical patterns are memorized and identified by dedicated microprocessors using similarity analysis for pattern recognition. The microprocessor identifies the species of the object gas and then calculates the concentration by the magnitude of sensor output [15].

A more general expression of this approach has been proposed and is shown in Figure 11. Multiple sensors $S_1, S_2, \ldots S_i$ have been designated for different gas species $X_1, X_2, \ldots X_N$. Matrix Q_{ij} describes multiple sensor sensitivities for gas species. It expresses selectivity and cross–sensitivity of the sensors. If all sensors have a unique selectivity for a certain species, all matrix components except diagonal ones are zero. However, gas sensors have imperfect selectivity and cross–sensitivity for multiple gas species; therefore, a microprocessor identifies an object species using the pattern recognition algorithm [16]. In this approach, imperfect selectivity of sensor materials is enhanced by electronic signal processing of microprocessor.

4.3 Approach using functional mechanical structure

If the signal processing function is implemented in the geometrical or mechanical structure of the sensors themselves, processing of the signal is simplified and a rapid response can be expected.

An optical information processing relating to image processing is discussed as an example.

Figure 12 shows a configuration for a coherent image processing system. Three convex lenses (focal length:f) are installed in coaxial arrangement. This system performs processing of Fourier transform and correlation calculation for spatial pattern recognition or image feature extraction.

Coherent light from the point source is collimated by lens L_1. The input image to be processed is inserted as a space varying amplitude transmittance $g(x_1, y_1)$ in plane P_1. Lens L_2 transforms g, producing an amplitude distribution $k_1G(x_2/\lambda f, y_2/\lambda f)$ across P_2, where G is the Fourier transform of g, and k_1 is a complex constant and λ is wave length of the light. A filter h is inserted in this plane P_2 to modulate the amplitude and phase of the spectrum G. If H represents the Fourier transform of h, then the amplitude transmittance of spatial frequency filter τ should be

$$\tau(x_2, y_2) = k_2H(x_2/\lambda f, y_2/\lambda f) \qquad (6)$$

The amplitude distribution behind the filter is proportional to GH. Finally, the lens L_3 transforms this amplitude distribution to yield an intensity distribution $I(x_3, y_3)$ on P_3.

$$I(x_3, y_3) = K\left|\iint_D g(\varepsilon, \eta)\bullet h(x_3-\varepsilon, y_3-\eta)d\varepsilon d\eta\right|^2 \qquad (7)$$

Reversal of the co-ordinate system on the plane P_3 is introduced to remove the sign reversal due to two Fourier transformations [17].

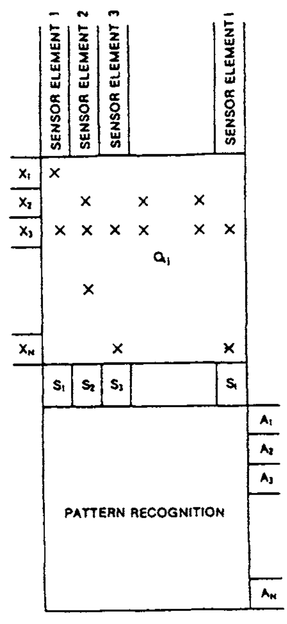

Figure 11 Gas analysis using multiple gas sensors and pattern recognition

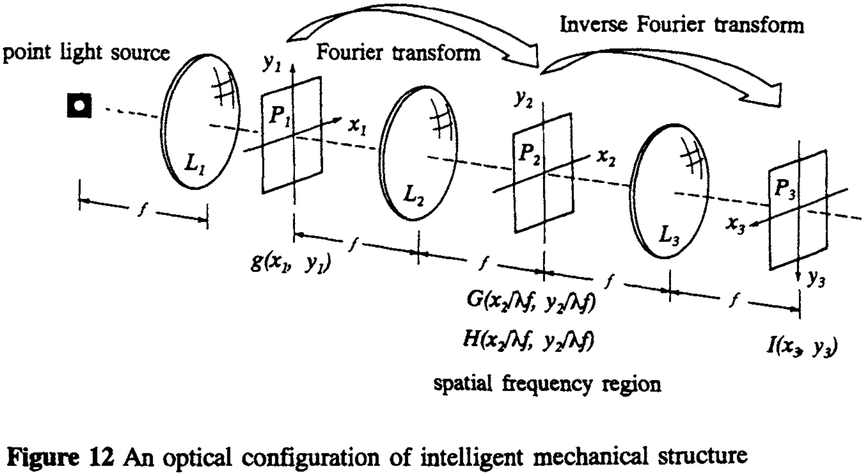

Figure 12 An optical configuration of intelligent mechanical structure

If we use filter h with suitable characteristics, we can extract useful feature from input image g by the filtering in spatial frequency region. As seen eq. (7) the output image on P_3 is proportional to two dimensional correlation of g and h. Computing of correlation between two figures can be carried out instantaneously utilizing Fourier transforming properties of lenses. This intelligent structure can be applicable for pattern recognition.

Another example is man's ability to tell three-dimensional directions of sound sources with two ears. We can also identify the direction of sources even in the median plane.
The identification seems to be made based on the directional dependency of pinna tragus responses. Obtained impulse responses are shown in Figure 13. The signals are picked up by a small electret microphone inserted in the external ear canal. Sparks of electric discharge are used as the impulse sound source. Differences in responses can be easily observed.
Usually at least three sensors are necessary for the identification of three-dimensional localization. Therefore, pinnae are supposed to act as a kind of signal processing hardware with inherent special shapes. We are studying this mechanism using synthesized sounds which are made by convolutions of impulse responses and na ural sound and noise [18].
Not only human ear systems, but also sensory systems of man and animals are good examples of intelligent sensing systems with functional structure.
The most important feature of such intelligent sensing systems is integration of multiple functions: sensing and signal processing, sensing and actuating, signal processing and signal transmission. Our fingers are typical examples of the integration of sensors and actuators. Signal processing for noise rejection, such as lateral inhibition, is carried out in the signal transmission process in the neural network.

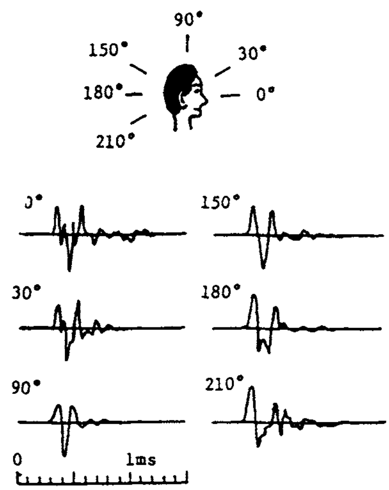

Figure 13 Pinna impulse responses for sound source in the median plane

4.4 Future image of intelligent sensors [7]

Intelligent sensing system on a chip
The rapid progress in LSI circuit technologies has shown a highly integrated advanced sensing system with learning ability. Neural networks and optical parallel processing approaches will be the most reasonable to overcome adaptability and wiring limitations.
An optical learning neurochip integrated on a single chip device has been proposed. The device is developed for pattern recognition and has a multilayer neural network structure with an optical connection between the layers. A number of photoemitter devices are arrayed on the top layer, while the signal transmission layer is built in the second layer. A template for an object pattern or special pattern memories is in the third, computing devices are in the fourth, and power supplies are in the bottom layer.

Image processing, such as feature extraction and edge enhancement, can be performed in the three-dimensional multifunctional structure. The pattern recognition function is formed by the neural network configuration and parallel signal processing. As previously described in this paper, the important feature of sensing systems of man and animals is adaptability. The feature is formed by such integration of multiple functions and distributed signal processing.

It is important to note that the approach to the future image is not single but three fold, each of which should be equally considered. In this book, these approaches are discussed with typical examples for various purposes. It is also important to note that a hierarchical structure is essential for advanced information processing and different roles should be reasonably allocated to each layer.

5. FUTURE TASKS OF INTELLIGENT SENSORS

This section will discuss what is expected of the intelligence of sensors in future. The author considers the following three tasks to be the most important examples to meet social needs.

5.1 Fault detection and forecast using machine intelligence

High performance or a highly efficient system is very useful if it works correctly, while it is very dangerous, if it does not. It is important to detect or forecast trouble before it becomes serious.

As I previously pointed out in the beginning of this chapter, a well-defined model of the abnormal state has not yet been realized. The present sensor technology is weak with regard to abnormal detection. Sensor fusion is a possible countermeasure to overcome the difficulty. At the moment the problem has been left unsolved for improved machine intelligence combining sensed information and knowledge.

5.2 Remote sensing of the target composition analysis

Chemical composition analysis is mostly carried out on the basis of sampled substance. In some cases, sampling of object material itself is difficult. For example, ozone concentration in the stratosphere is space- and time-dependent, but it is also a very important item to know for our environmental safety.

Remote sensing is indispensable for monitoring the concentration and distributions. Spectrometry combined with radar or lider technologies may be a possible approach for the remote sensing of ozone concentration.

Another important area of remote composition analysis is noninvasive biomedical analysis. In the clinical or diagnostic analysis, noninvasive composition analysis of human blood is very useful, if the analysis is reliable.

Composition analysis without sampling is easily influenced by various noise or media in between the sensing system and the object ingredient. Machine intelligence of the sensing system is expected to solve the problems.

5.3 Sensor intelligence for efficient recycle of resources

Modern production systems have realized an efficient and automatized production from source material to products. However, when the product is not in use or abandoned, the recycling process is not efficient nor automatized. The costs for recycling reusable resources are expensive and paid for by society. If the recycling of reusable resources is done efficiently and automatically, it can prevent contamination of our environment and a shortage of resources. Life cycle resource management will then be realized, as shown in Figure 14.

A sensing system for target material or components is indispensable for an automatic and efficient recycling process.

For this sensing system machine intelligence is necessary to identify the target ingredient or a certain component.

This is very important task of the intelligent sensing system.

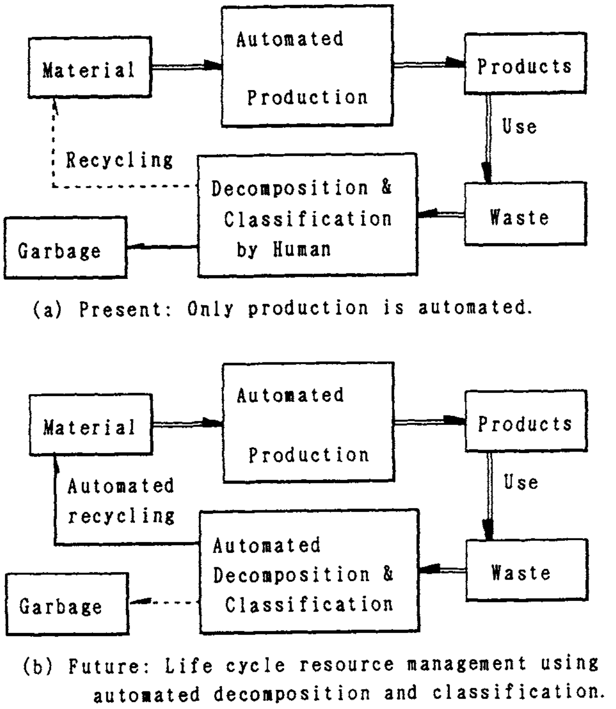

Figure 14 Advanced resource recycling process using intelligent sensing systems

REFERENCES

1 W.H. Ko & C.D. Fung, VLSI and Intelligent Transducers, Sensors & Actuators Vol.2(1982) pp.239.
2 S. Middelhoek & A.C. Hoogerwerf, Smart Sensors: When and Where?. Sensors & Actuators Vol.8(1985) pp.39.
3 J.M. Giachino, Smart Sensors, Sensors & Actuators Vol.10 (1986) pp.239.
4 K.D. Wise, Integrated Sensors; Interfacing Electronics to a Non electronic World, Sensors & Actuators Vol.2 (1982) pp.229.
5 J.E. Brignell, Sensors in Distributed Instrumentaion Systems, Sensors & Actuators, Vol.10 (1986) pp.249.
6 H. Yamasaki, Sensors and Intelligent systems, Y. Hashimoto and W. Day ed. Mathematical and Control Applications in Agriculture and Horticulture (Proc. IFAC/ISHS Workshop Matsuyama 1991) pp.349, Pergamon Press.
7 H. Yamasaki, Future Tasks for Measurement Technologies, Journal of The Society of Instrument and Control Engineers (SICE) Vol.31 (1992) pp.925 (in Japanese).
8 H. Yamasaki and K. Takahashi, An Intelligent Sound Sensing System Using Sensor Fusion, Digest of Technical Papers Transducers'93 pp 2 1993.
9 K. Takahashi and H. Yamasaki: Self-adapting Microphone System, S. Middelhoek ed. Proc. of Transducers'89 Vol.2 pp.610 Elsevier Sequoia S.A. 1990.
10 H. Yamasaki: Approaches to Intelligent Sensors, Proc. of the 4th Sensor Symposium, pp.69,(1984).
11 J. Van der Spiegel, Computational Sensors of the 21st Century, Extended Abstracts of International Symposium on Sensors in the 21st Century (1992 Tokyo) pp.51.
12 Mackintosh International, Sensors, Vol.24 (1981).
13 Digital smart transmitter DSTJ 3000, Journal of SICE 22(12) pp.1054 (1983)(in Japanese).
14 K. Ikeda et al., Silicon Pressure Sensor with Resonant Strain Gauges Built into Diaphragm, Technical Digest of the 7th Sensor Symposium, Tokyo,(1988), pp.55.
15 A. Ikegami and M. Kaneyasu: Olfactory Detection Using Integrated Sensor, Digest of Technical papers for Transducers'85 pp.136 (1985).
16 R. Muller and E. Lange, Multidimensional Sensor for Gas Analysis, Sensor and Actuators Vol.9 (1986), pp.39.
17 J. Goodman, Introduction To Fourier Optics, McGraw-Hill Physical and Quantum Electronics Series (1968).
18 Y. Hiranaka & H. Yamasaki:Envelope Representations of Pinna Impulse Responses Relating to 3-dimensional Localization of Sound Sources, J. Acoustical Society of America, Vol.73(1983), pp.291.